

Program 3

Code a dockerfile on python-flask or Node js Application

File Structure

- Dockerfile
- [index.js](#)
- package.json

Dockerfile

#Program for nodejs application in docker

#Created by : B M MADHUCHANDRA

Use Node official image
FROM node:20-alpine

Create app directory
WORKDIR /app

Copy package.json and install dependencies
COPY package*.json ./
RUN npm install

Copy rest of the app
COPY . .

Expose port
EXPOSE 3000

Start the app
CMD ["npm", "start"]

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index.js

```
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3000;

app.get("/", (req, res) => {
  res.send("Hello from Dockerized Node.js App ");
});

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

Package.json

```
{
  "name": "prg3",
  "version": "1.0.0",
  "description": "Simple Node.js app in Docker",
  "main": "index.js",
  "scripts": {
    "start": "node index.js"
  },
  "author": "B M MadhuChandra",
  "license": "ISC",
  "dependencies": {
    "express": "^4.19.2"
  }
}
```

Commands

Npm install

docker build -t program3:1.1 .

docker run -it -p 3000:3000 program3:1.1

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```
_bash ➤ prg3 14ms
sudo nano Dockerfile
[sudo] password for tantravruksha:
_bash ➤ prg3 27s 824ms
nano package.json
_bash ➤ prg3 17s 58ms
npm install

added 69 packages, and audited 70 packages in 3s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
_bash ➤ prg3 2s 803ms
nano index.js
_bash ➤ prg3 10s 340ms
docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
program3             1.1                12f6fbff37fe       3 minutes ago      134MB
bmmadhuchandra/firstimage 1.1                decfdab3d00d       7 days ago         147MB
firstimage           1.1                decfdab3d00d       7 days ago         147MB
_bash ➤ prg3 41ms
docker rmi -ff 12f dec
Untagged: program3:1.1
Deleted: sha256:12f6fbff37fe5cd0e307d5811afef6a331cc18644ee2d889d0fa7d37f8b879d3
Untagged: bmmadhuchandra/firstimage:1.1
Untagged: bmmadhuchandra/firstimage@sha256:2c1b36fd128b1a6737da2c38f11b390beddfa57f00cf567f5a8dde66b8dd4b80
Untagged: firstimage:1.1
Deleted: sha256:decfdab3d00d0bcf506ce155026c4c439dffabb65b15c85c3c459346fd07ccb2
_bash ➤ prg3 49ms
docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
_bash ➤ prg3 33ms
docker build -t nodeapp:1.1 .
[+] Building 4.5s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 397B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> [internal] load build context
=> => transferring context: 2.31MB
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install --only=production
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:e709df02559f72b5e1bb87a3b8a9e1d551fea306bc06d7811cd14d02823e0c5c
=> => naming to docker.io/library/nodeapp:1.1
_bash ➤ prg3 4s 656ms
```

```
_bash ➤ prg3 35ms
>> docker run -it -p 3000:3000 nodeapp:1.1
```

```
> prg3@1.0.0 start
> node index.js
```

Server running on port 3000

^C

